CHAPTER 3

# Function limits

S21, S23, S24, S26

**Before you start · Function limits**

Without a calculator, in six minutes. Answer briefly. Mobilize the sign, factorization and the then choose the targeted pick-up indicated under the questions.

1. Factoring $x^2 - 4$.

2. Give the sign of $\frac{x-1}{x+2}$ on $]-2;1[$.

3. Simplify $\sqrt{x^2}$ for $x < 0$.

4. What is the limit of the sequence $\frac{1}{n+1}$ ?

5. Determine the dominant term of 3 $x^3 - x$.

6. To give a prohibited value for $\frac{1}{x-2}$.

**Responses**

$(x-2)(x+2)$ ; negative ; $-x$ ; 0 ; $3x^3$ ; 2.

**Targeted review after the diagnostic check**

Repeat only the missed questions. For each question: **1.** Develop (x-2)(x+2) to check. **2.** Make two lines of signs and then quotient them. **3.** Test x=-3 and then generalize; R28. **4.** Resuming the unit limits of sequences and band around 0. **5.** Compare the degrees before any simplification. **6.** Cancel denominator before calculation. Questions R serve as support with the teacher, even if their collective ritual is planned later.

**Objectives**

Calculating limits at one point and infinitely; lifting an indetermination; interpreting a Asymptote.

## Course notes

### Sense of a limit

$f(x) \to \ell$ where $x \to +\infty$ means that $f(x)$ can be made as close as desired to $\ell$ for all $x$ large enough. $a$, $x$ of the estate, close to $a$ and distinct from $a$. The limits to the left and right are marked $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. When the domain has points on both sides of $a$a bilateral limit exists if and only if Two lateral limits exist and are equal. At one end of the domain, the limit of the only accessible side.

**Graphical interpretation**

If $f(x) \to \ell \in \mathbb{R}$ where $x \to +\infty$ or $-\infty$, right $y = \ell$ is a horizontal asymptote in that direction. $f(x) \to \pm\infty$ where $x \to a$ one side, the right $x = a$ is an asymptote vertical.

A curve can cut a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

### References and operations

In $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. In $-\infty$, the sign $x^m$ depends on the parity of $m$. The rules of sum, product, quotient and comparison are those of the sequences, with the same forms Indeterminate.

**Example.** For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. Limitations in $\pm\infty$ 2 in 3 $^-$, $f \to -\infty$ ; in $3^+$, $f \to +\infty$. Asymptotes are $x = 3$ and $y = 2$.

**Maritalization.** For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, attention: $\sqrt{x^2} = |x| = -x$.

### Exponential and Comparative Growth

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

Deduced from it $x^m e^{-x} \to 0$ in $+\infty$.

**Demonstration of the limits of the exponential.** The study of $e^x - x - 1$ given $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ in $+\infty$. By posing $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

**Demonstration of comparative growth.** For the integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If inequality is acquired by order $k$, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value in 0 is positive. Heredity follows. By choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

**Choose a transformation**

Polynomial quotient: factor the greatest powers. Root difference: quantity
quotient near a zero: factorize and study the signs.
power: compared growth, after variable change if necessary.

### Comparison and composition: conditions of employment

If $g(x) \leq f(x) \leq h(x)$ near the point or infinity studied, and if $g, h$ have the same real limit $\ell$, then $f(x) \to \ell$ (Theorem of the gendarmes) $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit $-\infty$, on uses an increase by a function tending to $-\infty$. If $u(x) \to a$ and $v$ is continuous in $a$, then $v(u(x)) \to v(a)$, when the compound is defined. For example $\sqrt{1+\frac{1}{x}} \to 1$ in $+\infty$. For a limit infinite of the external function, check which side $u(x)$ Approach the bound: $\frac{1}{u(x)}$ Not the same behaviour according to the sign of $u(x)$.

# Read a limit, choose a transformation

## 3.1 The approach side is part of the question

First use: S23.

**Example Solved · Vertical Asymptote**

For $f(x) = 1 + \frac{1}{x-2}$, the domain is $\mathbb{R} \setminus \{2\}$. When $x \to 2^-$ the denominator is negative and tend towards 0, so $f(x) \to -\infty$. When $x \to 2^+$, $f(x) \to +\infty$. The right $x = 2$ is asymptote vertical; no value $f(2)$ exists. Infinity, $f(x) \to 1$.

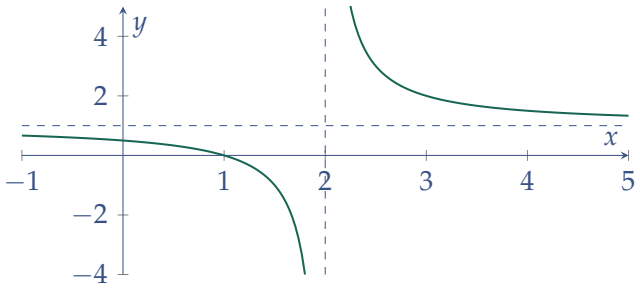



**Figure 3.1.** Find the two lateral boundaries on the drawing: the branches are not connected.

**A1 · Immediate application** Determining the limits of $g(x) = 3 - \frac{2}{x+1}$ in $-1^-$, $-1^+$, $-\infty$ and $+\infty$. Indicate asymptotes.

## 3.2 Undetermined form requires calculation

First use: S24.

**Example Solved · The sign hidden in a root**

For $x < 0$, $\sqrt{x^2} = |x| = -x$. Thus, in $-\infty$,

$$\frac{\sqrt{x^2+1}}{x} = \frac{|x|\sqrt{1+1/x^2}}{x} = -\sqrt{1+1/x^2} \longrightarrow -1.$$

In $+\infty$, the same expression tends to 1. Omitting the absolute value would produce a limit positive for a negative expression.

**A2 · Guidance which disappears** Calculate in $+\infty$ the limits of $\frac{2x^2+1}{x^2-x}$, $\sqrt{x^2+4x} - x$ and $x\left(\sqrt{1+1/x} - 1\right)$. For the first, it is suggested to divide by $x^2$ ; choose alone for others.

**A3 · Build Examples** Build three limit quotients of form $0/0$ in 0, of which the limits are 0, 1 and $+\infty$. Specify the approach side if necessary.

### 3.3 Comparative growth and variable change

First use: S26; logarithm after S49.

**Ownership · exponential and power · course results**

For the integer $m \geq 1$, $e^x/x^m \to +\infty$ and $x^m e^{-x} \to 0$ in $+\infty$. To transport these results, identify the new variable and its limit.

**Example Solved · Carrying a Result**

When $x \to -\infty$, pose $t = -2x$ gives $t \to +\infty$ and $x^2e^{2x} = \frac{1}{4}t^2e^{-t} \to 0$. After S49, substitution $t = 1/x$ in $x \ln x$ for $x \to 0^+$ gives $-(\ln t)/t \to 0$. Logarithm n
Not before his study.

**A4 · Deferred application** After S26, calculate $\lim_{x\to+\infty}(x^2+3x)e^{-x}$ and $\lim_{x\to-\infty} xe^x$. To say why formally replace $x$ Infinity does not constitute justification.

### 3.4 An asymptote may be crossed

First use: S24.

**Definition · Horizontal Asymptote**

If $f(x) - \ell \to 0$ to one of the infinite, $y = \ell$ is an asymptote to this infinity.
a vertical distance that tends to 0; it does not prohibit any intersection at a finite distance.

**Example Solved · Two Behaviours**

On $\mathbb{R}$, $p(x) = 1 + 1/(1+x^2)$ remains strictly above $y = 1$. The function $q(x) = 1 + x/(1+x^2)$ cross that same right in $x = 0$, while tending towards 1 to the two infinites.
For $q$, the sign $q(x) - 1$ is the $x$.

**A5 · Refute and Explain** A student says that the curve of $q$ Can't cut his asymptote.
the coordinates of the point of intersection and the signs on both sides. Is it compatible with the Two limits?

**A6 · Choice of strategy** Determine $a, b$ so that $h(x) = \frac{ax+b}{x-1}$ have for horizontal asymptote $y = 2$ and verifies $h(0) = 3$. Then study the lateral limits in 1 and the position of the curve in relation to $y = 2$.

**Wait a minute at the courthouse.** For each of the expressions of the "Methods" application, announce first
This prevents us from concluding with the boundary operations and then explaining the role of the transformation.

**Review postponed · S49 and then S81** Resume variable change with $\ln x/x$ ; reclassify limits by mixing logarithms, exponentials and roots.

## Research activity

### A curve close to a right · 30 min.

The function is considered $f$ defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. We note $\mathcal{C}_f$ its curve in a coordinate system of the plan.

1. Show that, for all $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.

2. Determine the limits of $f$ in $-\infty$, $1^-$, $1^+$ and in $+\infty$.

3. Study the sign of $f(x) - 3$, then specify the position of $\mathcal{C}_f$ in relation to the right of equation $y = 3$.

4. Determining a Real $A > 1$ such as, for all $x > A$, $|f(x) - 3| < 0{,}01$.

5. The Curve $\mathcal{C}_f$ Does it cut its horizontal asymptote? Justify and explain why a window too wide graphic can give a misleading impression.

## Progressive exercises

**E1 · Direct application.** Determine $\lim\limits_{x \to +\infty} \dfrac{5x^3 - x}{2x^3 + 7}$ justifying.

**E2 · Application and justification.** Let $f$ the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits in $2^-$ in $2^+$, in $-\infty$ and in $+\infty$.

**E3 · Method to be organised.** Determine $\lim\limits_{x \to 1} \dfrac{x^2 - 1}{x - 1}$ specifying why the simplification used is valid in the vicinity of 1.

**E4 · Choice of strategy.** Determine $\lim\limits_{x \to +\infty} x^3 e^{-2x}$. Cite the comparative growth result used.

**E5 · Synthesis or demonstration.** Determine $\lim\limits_{x \to -\infty} \left(\sqrt{x^2 + 1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.

## Work in the manual or annal

**Barbazo, p. 96, Exercises 26 and 29.** In S23: Exercise 29, to connect boundaries and asymptotes. At home After S24: Exercise 26, to construct three counter-examples to an abusive calculation rule on limits.

## Duty at home

**DM3 · A family of curves · 45 min.**

For every real $a$, the function is considered $f_a$ defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. We note $\mathcal{C}_a$ its curve representative.

1. Determine the definition set of $f_a$, then its limits in $-\infty$ and in $+\infty$.

2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of $a$, specify the $\mathcal{C}_a$ in relation to the right of equation $y = a$.

3. Studying variations in $f_a$ distinguishing cases $a < 1$, $a = 1$ and $a > 1$. Clarification of possible extremema and where they are affected.

4. Take $a = 3$. Resolve, for $x \geq 0$, the inequitable $|f_3(x) - 3| \leq 0{,}02$.

## QCM A · lateral boundaries and transformations

After S24; QCM B after S26.

For each question, only one proposal is accurate.
consult the correction.

**1.** $\lim_{x\to 2^-} \frac{1}{x-2}$ is valid for:

A. $+\infty$ B. $-\infty$

C. 0 D. 2

**2.** For $x < 0$, $\sqrt{x^2}$ is valid for:

A. $x$ B. $x^2$

C. $-x$ D. $\pm x$

**3.** $\lim_{x\to+\infty} \frac{3x^2-x}{x^2+4}$ is valid for:

A. 0 B. $+\infty$

C. $-1$ D. 3

**4.** $\lim_{x\to 1} \frac{x^2-1}{x-1}$ is valid for:

A. 2 B. 0

C. $+\infty$ D. Does not exist

**5.** For $f(x) = 1 + \frac{\sin x}{x}$, $x > 0$, the right $y = 1$ :

A. is not asymptote B. is asymptote without intersection

C. is asymptote crossed an infinity of times D. is tangent at all points

## QCM B · Comparative growth and tool choice

For each question, only one proposal is accurate.
consult the correction.

**6.** $\lim_{x\to+\infty} x^2e^{-x}$ is valid for:

A. $+\infty$ B. 0

C. 1 D. $-\infty$

**7.** $\lim_{x\to-\infty} xe^x$ is valid for:

A. 0 B. $+\infty$

C. $-\infty$ D. 1

**8.** The appropriate tool for $\sqrt{x^2+2x} - x$ in $+\infty$ is:

A. the binomial B. VIT

C. a diversion D. Conjugation

**9.** If $f(x) \to 0$ and $g(x) \to +\infty$the product:

A. Always tends to 0 B. Always tend towards $+\infty$

C. requires an additional study D. Is impossible

**10.** If $f(x) \geq x^2$ for $x \geq 1$, then:

A. $f(x)/x^2 \to 1$ B. $f(x) \to +\infty$

C. $f$ is increasing D. $f(x) - x^2 \to 0$

**True or false · expected justification.** For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. A function never cuts its horizontal asymptote.

2. A finite limit in $a$ requires that $f(a)$ be defined.

3. The symbol $\infty/\infty$ gives a limit equal to 1.

## Corrected MCQs

1. **B.** The denominator tends to 0 by negative values. The side of the approach determines the sign.

2. **C.** The square root is positive or zero; $\sqrt{x^2} = |x| = -x$ for $x < 0$. This is not an equation to Two solutions.

3. **D.** Divide by $x^2$ gives $(3 - 1/x)/(1 + 4/x^2) \to 3$.

4. **A.** For $x \neq 1$, the quotient is $x + 1$. A prohibited value does not make a limit impossible.

5. **C.** The gap tends to be 0 and cancels in $k\pi$, $k \geq 1$, with change of sign. An asymptote may to be crossed.

6. **B.** It's $x^2/e^x$, which tends to 0 by comparison of growth rates. The exponential dominates every power fixed.

7. **A.** Lay down $t = -x \to +\infty$ : the expression is $-t/e^t \to 0$.

8. **D.** Multiply by the conjugate quantity gives 2/ $(\sqrt{1 + 2/x} + 1) \to 1$.

9. **C.** With $g = x$, take $f = 1/x^2$, $1/x$ or $1/\sqrt{x}$ : 0, 1 and $+\infty$.

10. **B.** The comparison theorem is enough for the limit. 2 $x^2$ refutes the two limits equality; inequality Don't give monotonicity.

**True/false.** False: Question 5. False: Question 4. False: $x/x \to 1$, $x/x^2 \to 0$, $x^2/x \to +\infty$.

## Corrected applications and support

**Application 1.** In $-1^{-}$ : $+\infty$ ; in $-1^{+}$ : $-\infty$ ; both infinites : 3. Asymptotes $x = -1$ and $y = 3$.

**Application 2.** Limitations: 2; $\frac{4x}{\sqrt{x^2+4x}+x} = \frac{4}{\sqrt{1+4/x}+1} \to 2$ ; $\frac{1}{\sqrt{1+1/x}+1} \to 1/2$. The expressions are: defined for $x$ big enough.

**Implementation 3.** For example $x^2/x \to 0$, $x/x \to 1$, $x/x^2 = 1/x \to +\infty$ in $0^{+}$. The functions are considered outside 0.

**Implementation 4.** The first tends to 0 by linearity and comparison of growth rates. For the second, $t = -x \to +\infty$ gives $xe^{x} = -te^{-t} \to 0$. The symbol $\infty$ is not a real in which one evaluates a function.

**Application 5.** $q(0) = 1$. $q(x) - 1$ is negative for $x < 0$, positive for $x > 0$. The limits relate to: the infinite ends of the domain, so do not contradict this intersection.

**Implementation 6.** The limit to infinity is $a$, therefore $a = 2$ ; $h(0) = -b = 3$, so $b = -3$. Thus $h(x) = 2 - 1/(x-1)$ : limits $+\infty$ in $1^{-}$, $-\infty$ in $1^{+}$ ; curve above 2 left of 1 and in Under the right.

**Reading then calculation**

Before any table, make the quotient sign anticipate. A numerical result contrary to the sign Expected must trigger a check. $p$ and $q$ on the same window to contest The image of an "inconstructible barrier".

## Conduct and correction

**Benchmarks for the teacher**

In S21, connect each limit to a graphical reading. In S23, require the signs of the denominator left and right. In S24, have the transformation chosen before calculating: factorization, conjugation, local development not used. In S26, separate demonstration from growths Oblique asymptotes are not required here.

### Corrected activity

1. Like 3 $x-2=3(x-1)+1$, for $x\neq 1$, $f(x)=3+\frac{1}{x-1}$.

2. In $\pm\infty$, $\frac{1}{x-1}\to 0$: the limits are 3 in 1 $^-$, $x-1\to 0^-$, therefore $f(x)\to-\infty$ ; in 1 $^+$, $x-1\to 0^+$, so $f(x)\to+\infty$.

3. The sign of $f(x)-3=\frac{1}{x-1}$ is negative for $x<1$, positive for $x>1$. The curve is therefore below and above asymptote, respectively $y=3$.

4. For $x>1$, $|f(x)-3|<0.01$ is equivalent to $\frac{1}{x-1}<0.01$, then $x>101$. We can choose $A=101$.

5. No: $f(x)-3=\frac{1}{x-1}$ The curve may appear to be confused with the asymptote at the precision of the screen, but it does not meet it.

**E1.** For $x\neq 0$, the quotient $\frac{5-\frac{1}{x^2}}{2+\frac{7}{x^3}}$. In $+\infty$, the inverse powers tend to 0 and the denominator to 2 $\neq 0$. The limit is $\frac{5}{2}$.

**E2.** In 2, $(x-2)^2\to 0$ by strictly positive values on both sides; its inverse therefore tends towards $+\infty$. In $\pm\infty$, the square tends towards $+\infty$ and its inverse towards 0.

**E3.** For $x\neq 1$, $x^2-1=(x-1)(x+1)$So the quotient is equal to $x+1$. Its limit in 1 is therefore 2. This simplification calculates the limit; it does not render the initial function defined in 1.

**E4.** Poisons $t=2x\to+\infty$. Then $x^3e^{-2x}=\frac{t^3}{8e^t}\to 0$, by comparative growth of the exponential and the power.

**E5.** The conjugate gives $\sqrt{x^2+1}+x=\frac{1}{\sqrt{x^2+1}-x}$. Where $x\to-\infty$, both terms $\sqrt{x^2+1}$ and $-x$ tend to $+\infty$ ; the denominator therefore tends to $+\infty$. The limit is 0, by positive values.

### Corrected from DM3

1. Like $x^2+1>0$ on $\mathbb{R}$, the domain is $\mathbb{R}$. By dividing by $x^2$ for $x\neq 0$, $f_a(x)=\frac{a+\frac{1}{x^2}}{1+\frac{1}{x^2}}\to a$ in $\pm\infty$.

2. $f_a(x)-a=\frac{ax^2+1-a(x^2+1)}{x^2+1}=\frac{1-a}{x^2+1}$. For $a<1$, the curve is above $y=a$ ; for $a>1$, it is below; for $a=1$, it is confused with this line.

3. By quotient, $f_a'(x)=\frac{[2ax(x^2+1)-2x(ax^2+1)]}{(x^2+1)^2}=\frac{2x(a-1}{(x^2+1)^2}$. If $a>1$, the sign is that of $x$ : decrease on $]-\infty,0]$, growth on $[0,+\infty[$ ; minimum $f_a(0)=1$ and no maximum (the bound $a$ is not (a) If $a<1$, the variations are reversed: maximum 1 and no minimum. $a=1$, the function is constant, value 1.

4. For $a=3$ and $x\geq 0$, inequality becomes $\frac{2}{x^2+1}\leq 0{,}02$. The denominator is positive, so it equals to $x^2+1\geq 100$, i.e. $x\geq\overline{99}$.

### Corrected work in the manual or an anal

**Exercise 26.** When $x \to +\infty$, functions $f(x) = x$ and $g(x) = x$ both tend towards $+\infty$, But their difference is no good. $f(x) = 2x$, $g(x) = x$, the difference tends to $+\infty$ ; by exchanging $f$ and $g$, it tends towards $-\infty$. These examples establish the indeterminate nature of the difference.

**Fiscal year 29.** The denominator cancels in $\frac{3}{2}$, therefore $D_f = \mathbb{R} \setminus \{\frac{3}{2}\}$. We're writing

$$f(x) = \frac{5 - 2x}{2x - 3} = -1 + \frac{2}{2x - 3}.$$

So $f(x) \to -1$ to the two infinites. $\frac{3}{2}$, the numerator of the second term is positive and its denominator tend towards 0: the limit on the left is $-\infty$, the right is $+\infty$. Asymptotes have as equations $y = -1$ and $x = \frac{3}{2}$.